

Replica-Based Moving Target Defense Against Injection Attacks in Software-Defined Industrial Control Systems

Xabier Etxezarreta¹, Federico Turrin², Iñaki Garitano¹, Mikel Iturbe¹, Urko Zurutuza¹,
and Mauro Conti³, *Fellow, IEEE*

Abstract—Recent incidents have demonstrated the increasing vulnerability of Industrial Control Systems (ICSs) to sophisticated and targeted attacks orchestrated by adversaries with high motivation, resources, and domain knowledge. Among these threats, False Data Injection (FDI) attacks have emerged as one of the main security threats to ICSs, involving the deliberate manipulation or injection of false data into the control system to deceive or disrupt operations. FDI attacks pose a significant risk due to their high capacity of concealment and ability to evade intrusion detection systems that rely on accurate ICS models. In this paper, we present DEFCLON, a novel Software-Defined Networking (SDN)-based Moving Target Defense (MTD) approach against FDI attacks. DEFCLON proactively replicates network packets across multiple network paths and adaptively selects a single path using a signaling game model to reach the destination end-device. We demonstrate the effectiveness of our approach through simulations, numerical analysis, and experiments on ICS network traffic and topologies. Experimental results show that DEFCLON is able to not only mitigate the effects of FDI attacks, but also to introduce different levels of uncertainty without degrading network performance, significantly increasing the difficulty for adversaries to gather information and launch attacks.

Index Terms—Moving target defense (MTD), industrial control systems, software-defined networking (SDN), game theory, injection attacks.

I. INTRODUCTION

INDUSTRIAL Control System (ICS) refers to a category of computerized systems used to monitor and control industrial processes in sectors such as manufacturing, energy,

water management, transportation, and more, including Critical Infrastructures (CIs) [1]. ICSs play a crucial role in automating and optimizing complex industrial operations by controlling and managing physical processes, machinery, and equipment. These systems typically consist of a combination of hardware, software, and network infrastructure. They are designed to collect data from various sensors and devices in real time, analyze that data, and initiate control actions to maintain optimal operation and safety within industrial facilities.

In the past, due to the critical nature of these systems, ICSs were traditionally isolated from other networks, ensuring a high level of security and stability [2]. However, with the advent of the IT-OT (Information Technology-Operational Technology) convergence, the landscape has changed, giving rise to new threats and challenges [3]. This convergence has brought numerous benefits, including increased efficiency, improved data analysis, and enhanced decision-making capabilities. However, it has also introduced vulnerabilities and exposed ICSs to new risks. As ICSs become more connected to IT networks and the Internet, they become more susceptible to malicious activities. Attackers now have a broader attack surface to target, and a successful breach of an ICS can have severe consequences, including operational disruptions, financial losses, and even safety hazards. Among the significant threats to ICSs, False Data Injection (FDI) attacks pose a particularly grave risk due to the growing connection to IT networks and the Internet. These attacks can involve malicious actors injecting false or manipulated data into the ICS traffic transmitted from the network, potentially resulting in severe consequences [4].

A major concern is the use of unencrypted industrial protocols to transmit data. Industrial protocols such as Modbus, DNP3, and OPC were originally designed for closed, isolated environments where security was not a primary concern. As a result, these protocols often lack encryption and authentication mechanisms, making them highly vulnerable to interception and manipulation by attackers. In addition, many ICSs operate in real time with strict latency constraints. For example, in power systems, IEC 61850 standards require fault isolation and protection services to meet a latency limit of 3 ms. However, the implementation of access control and cryptographic measures can introduce significant delays that exceed these latency requirements and disable essential real-time ICS operations [5]. Recent studies of real-world ICS traffic over the Internet have reported the

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Xabier Etxezarreta, Iñaki Garitano, Mikel Iturbe, and Urko Zurutuza are with Data Analytics and Cybersecurity (DANZ) research group, Electronics and Computing Department, Mondragon University, 20500 Arrasate-Mondragon, Spain (e-mail: xetxezarreta@mondragon.edu; igaritano@mondragon.edu; miturbe@mondragon.edu; uzurutuza@mondragon.edu).

Federico Turrin is with the Department of Mathematics, University of Padova, 35121 Padua, Italy, and also with SpritzMatter srl, 35121 Padua, Italy (e-mail: federico.turrin@spritzmatter.com).

Mauro Conti is with the Department of Mathematics, University of Padova, 35121 Padua, Italy, and also with Örebro University, 70182 Örebro, Sweden (e-mail: mauro.conti@unipd.it).

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vulnerabilities of these systems [6], [7], highlighting a lack of security features in their communications.

To address FDI attacks, researchers have employed fault diagnosis and bad data detection techniques (e.g., least squares method [8] or Kalman filter [9]). However, most of these detectors typically rely on an accurate ICS model, which affects the precision and reliability of intrusion detection, allowing an attacker that possesses knowledge of the ICS model to design stealthy FDI attacks to evade these intrusion detection systems [10]. In fact, one of the critical factors contributing to the success of these attacks in ICS is the ability of attackers to gain extensive knowledge about the topology and architecture of the targeted ICS during the reconnaissance phase. As adversaries accumulate deeper knowledge about the targeted system, their capacity to launch progressively more sophisticated, stealthy and impactful attacks escalates.

To invalidate the knowledge-gathering and stealth capabilities of attackers, Moving Target Defense (MTD) has been proposed to counter FDI attacks [11]. MTD is a dynamic security approach that aims to counteract the advantage gained by attackers through increasing system complexity and uncertainty. Rather than relying on the static nature of ICS, MTD introduces constant variability and unpredictability into the system's characteristics, making it significantly more challenging for adversaries to gather accurate information, hindering the attack design and implementation process. Nonetheless, deciding how often to apply MTD countermeasures poses a challenge in establishing a trade-off between defense performance and attack surface exposure time. Longer intervals can result in an extended window of vulnerability, allowing attackers to carry out their malicious activities. Conversely, short intervals can result in an inefficient allocation of defensive resources.

Therefore, it is particularly important to design an MTD approach against FDI attacks that dynamically adapts the attack surface to changing threat scenarios without the need for an ICS model. The dynamic aspect of MTD adds an extra layer of complexity, requiring network orchestration technologies and tools to implement these types of defenses. Software-Defined Networking (SDN) has become a recurring technology in many research works that propose the use of MTD for network security [12]. As defined in RFC 7426 [13], SDN is an approach to network programmability, meaning the ability to dynamically initialize, control, modify and manage network behavior through open interfaces. To achieve this, the control plane is separated and centralized in an external entity called SDN controller, while the data plane remains on the network devices, focusing on packet forwarding functionality. SDN introduces a number of advanced capabilities, including dynamic network configuration, centralized control, granular network traffic analysis, flexible policy enforcement, automated intrusion response, continuous monitoring and analysis, and real-time incident response. These capabilities are difficult to achieve using traditional networking technologies, but they are essential for the development of adaptive MTD. Currently, the integration of SDN in industrial environments is still in its early stages of development [14]. However, SDNs have already been successfully implemented in IT environments, and several studies have demonstrated

their effectiveness in developing security solutions for ICS, particularly in the attack detection and response field [14], [15], [16].

To this end, in this paper we propose DEFCLON, a novel SDN-enabled MTD framework to mitigate the impact of FDI attacks in ICS. Our focus is on creating proactive and adaptive network configurations that effectively change and adapt the existing attack surface, making it more difficult for adversaries to perform FDI attacks. Building upon previous research on MTD, we propose a multi-layered MTD strategy that is centrally implemented on top of the SDN controller. DEFCLON proactively replicates industrial network packets across different network paths, reducing the ability to intercept network traffic and limiting the ability of adversaries to gather industrial process knowledge. Among the different paths, DEFCLON incorporates an adaptive system based on a signaling game model [17], that ensures that paths exhibiting suspicious behavior are not selected for transmitting data to the destination end-device. Our results demonstrate that DEFCLON effectively mitigates FDI attacks in ICS. It successfully prevents attack traffic from reaching the destination device, even compared to a fully random/non-adaptive traffic replication strategy. In summary, the contributions of this paper are as follows:

- We propose a novel SDN-enabled MTD framework to mitigate FDI attacks. The framework replicates industrial network packets from a selection of different paths randomized over time. From the pool of selected paths to transmit packets, we select a single path to transmit data to the destination end-device, reducing attacker's ability to inject packets into the industrial process.
- To further enhance the effectiveness of the replication process, we propose a signaling game model implemented on top of the SDN controller. The game model analyzes packets transmitting from each path in real-time without relying on a specific ICS model and adaptively selects one path to send data to the intended destination device. This adaptive approach ensures that paths displaying unusual or suspicious behavior are not chosen for data forwarding.
- We employ a simulated SDN-enabled industrial test environment to evaluate the proposed defense approach. Through thorough numerical analysis and evaluation of the results obtained, we demonstrate the effectiveness of the proposed MTD framework in mitigating FDI attacks in ICS.

The rest of the paper is organized as follows. In Section II, we provide a discussion of related work and in Section III, we define the threat model. Section IV describes DEFCLON treating its theoretical aspects. Then, we present the experimental setup and results in Sections V and VI. We discuss the limitations of the proposed work in Section VII, and finally, we conclude this work on Section VIII.

II. RELATED WORK

In this section, we discuss and provide an overview of existing research and literature relevant to this work. First, we present research efforts on different MTD techniques. Second,

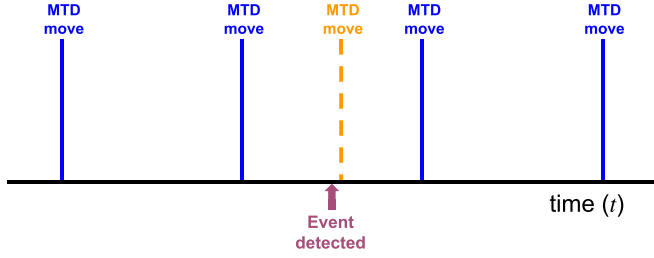


Fig. 1. Differences between proactive and adaptive MTD [18]. The blue vertical lines represent a proactive MTD strategy that is triggered at fixed time intervals, while the dashed orange vertical line represents an adaptive MTD strategy that is triggered by an event.

we provide insights into the applicability of signaling games for attack mitigation. Finally, we analyze different MTD techniques proposed in the literature against FDI attacks.

A. Moving Target Defense Techniques

The goal of MTD techniques is to enhance the security of computer systems and networks by constantly changing their attack surface and making them more challenging to exploit. Different MTD techniques can be classified according to their operational area [18].

Shuffling-based MTD techniques employ dynamic randomization of system configurations. This randomization process obfuscates the configurations, making it difficult for attackers to gather information during the reconnaissance phase. As a result, the attack surface is reduced, and the complexity of identifying and exploiting vulnerabilities is increased. Examples within this category include IP address randomization [19], port number randomization [20], network path/route randomization [21], and packet header randomization [22]. Furthermore, several proposals have been proposed that combine multiple shuffling-based techniques. For instance, authors in [23] propose a low-delay approach involving IP, MAC, and port numbers randomization to mitigate reconnaissance attacks in industrial networks.

Diversity-based MTD refers to the provision of equivalent services using different implementations, enhancing system resilience. Code diversity aims to divide a program into components that can be executed in different execution environments [24]. Software diversity techniques involve deploying alternative versions of web servers, applications, or virtual servers to enhance network resilience [25]. Furthermore, programming languages diversity helps to mitigate risks of SQL or code injection attacks [26].

Redundancy-based MTD involves deploying systems that offer the same functionality in order to enhance reliability and availability. As an example, authors in [27] propose network session redundancy in Cyber-Physical Systems (CPS) communications to combat traffic analysis attacks.

Hybrid MTD consists of combining shuffling, diversity and redundancy-based MTD techniques [28], [29].

A major challenge in proactive MTD solutions is determining the optimal timing for applying MTD countermeasures. Fig. 1 illustrates the difference between proactive and adaptive MTD

strategies. If the interval is too long in a proactive strategy, attackers have enough time to compromise the system. Conversely, if the interval is too short, MTD may be triggered unnecessarily, consuming defense resources. To overcome this, we propose a hybrid approach that combines shuffling and redundancy-based MTD with proactive and adaptive MTD mutations using a signaling game model. This approach allows operations to adapt to both normal and threat scenarios, without relying on a completely random approach.

B. Signaling Games in Cybersecurity

Game theory has emerged as a framework for understanding and analyzing strategic interactions between individuals or entities. It has found applications in various fields, and one area where game theory has gained significant relevance is cybersecurity [30], providing insights into the strategic decisions made by both attackers and defenders. One particular aspect of game theory that has proven valuable in this context is signaling games. These games involve players who have incomplete information and strategically send signals to influence the behavior of others. In the field of attack mitigation, signaling games provide a framework for modeling interactions between attackers and defenders, where defenders do not know exactly whether the signal received comes from a legitimate user or an attacker. Aydeger et al. [31] propose an adaptive SDN-enabled approach to mitigate DDoS attacks. Their method involves the randomization of network paths and employs a signaling game model to make informed decisions on when and where to apply this randomization. Priyadarsini et al. [32] present a framework for both mitigating and evaluating attacks targeting SDN controllers. Their approach uses a signaling game to model communications between switches and SDN controller. Signaling games have also proven to be a valuable tool for integrating Moving Target Defense (MTD) techniques with cyber deception strategies. Zhou et al. [33] introduce an approach that combines cyber deception with multiple MTD strategies, such as IP randomization, path randomization, and response time adaptation. Furthermore, they introduce a defender-led signaling game model, allowing for the proactive transmission of deceptive signals to attackers. This approach addresses the inherent information asymmetry in cyberattacks, giving defenders a strategic advantage.

C. MTD for FDI Attacks Mitigation

The effectiveness of FDI attacks in evading detection systems has been well established, and attackers with extensive knowledge of the parameters of the ICS model pose a significant threat. As a result, current MTD strategies for FDI attacks primarily focus on limiting the attacker's ability to gain an in-depth understanding of the system or detecting these threats by revealing attackers in MTD environments [34], [35].

A prominent MTD strategy in power systems involves altering the physical layer using Distributed Flexible AC Transmission System (D-FACTS) devices [36], [37] to invalidate an attacker's knowledge for launching stealthy FDI attacks. However, this approach has significant limitations, as its effectiveness depends

TABLE I
A COMPARATIVE ANALYSIS OF DIFFERENT MTD STRATEGIES AGAINST FDI ATTACKS

Reference	Method	Infrastructure Requirements	ICS Model Dependency	Adaptation Strategy	Key Limitations
[36]	Susceptance perturbations	D-FACTS devices installation	Yes	Proactive	High infrastructure cost, strict topological conditions
[37]	Reactance perturbations	D-FACTS devices installation	Yes	Proactive	Computational costs, models training, sensitive to the selection of hyperparameters
[39]	Topology switching	Spare busbars, switching equipment	Yes	Proactive	Switching actions limited by the physical layout of the power system (e.g., lines and busbars)
[40]	System reconfiguration and anomaly detection	Redundant actuators and sensors	Yes	Proactive and reactive	Performance vs. security tradeoff, requires component redundancy
[41]	Control/Masurement matrices perturbation	Actuators and sensors with adjustable gains, secure seed synchronization to ensure the control center and the field devices are using the same perturbed matrices	Yes	Proactive	Vulnerable to matrix estimation attacks, dependent on the common seeds used to generate the matrix perturbations
[42], [43]	Data replication and randomization	IoT devices for data replication, out-of-band private data exchange network	Yes (for optimal allocation)	Proactive	Probabilistic defense, reduces but does not eliminate attack impact due to its proactive nature
DEFCLON (our proposal)	Data replication and adaptive path selection	SDN infrastructure, redundant network paths	No	Proactive and adaptive	SDN adoption challenges, reliance on path redundancy

on strict topological conditions [36]. Due to these constraints and the high infrastructure costs of D-FACTS, subsequent research has focused on developing more cost-effective solutions by minimizing the number of required devices and their operational expenses, while also analyzing the defense's impact on system stability [38].

As an alternative to parameter perturbation, some researchers have proposed topology switching as an MTD mechanism [39]. This approach reconfigures the system topology, for instance through busbar switching, leveraging existing spare busbars to serve as a security measure without requiring extra investment. To manage the complexity of selecting optimal switching actions, the defense process was modeled as a Markov Decision Process (MDP) and solved using Deep Reinforcement Learning (DRL) to achieve fast and efficient decision-making suitable for large-scale systems.

For instance, in [40], a reactive and proactive MTD mechanism is proposed that dynamically changes the controller, altering cyber-physical system parameters, making it difficult for attackers to perform reconnaissance in the system and to detect compromised actuators and sensors. Similarly, the authors in [41] propose a framework for detecting Stuxnet-like attacks and limiting adversary knowledge by actively perturbing sensor and control commands. To counter the emergence of adversarial FDI attacks designed to bypass even advanced Machine Learning (ML)-based detectors, other approaches deploy a pool of diverse Deep Neural Network (DNN) models to reduce an attack's transferability, which was combined with physics-based perturbations to enhance detection accuracy [37]. Another strategy employs proactive data replication by integrating new IoT devices to duplicate control commands and sensor readings, thereby limiting an attacker's state estimation capabilities and reducing the likelihood of compromised data reaching its destination [42], [43].

However, existing approaches present several shortcomings, as many depend on accurate system models which can be difficult to maintain and scale. Furthermore, proactive MTD strategies that rely on fixed randomization frequencies struggle to adapt to evolving threats. To address these issues, we propose an SDN-enabled proactive and adaptive traffic replication framework to mitigate FDI attacks in the ICS domain. Our framework leverages the central management of SDN to distribute traffic across multiple paths and, without relying on an ICS model, employs a signaling game to detect suspicious activity. This allows the system to adaptively manage the path selection pool,

preventing paths with suspicious behavior from reaching the destination and overcoming the challenge of defining an optimal randomization frequency. Table I presents a comparative analysis of different MTD strategies against FDI attacks in ICS.

III. THREAT MODEL

For our threat model, we consider an ICS network that is centrally managed using an SDN controller. We define the attacker's characteristics and the attack process as follows:

Attackers' Capabilities: An attacker within an industrial network poses a significant threat to the integrity of industrial communications. The attacker's potential capabilities are:

- *Eavesdropping:* The attacker can intercept data packets transiting the network from any communication channel or path. This allows them to gather sensitive information or monitor system behavior. There are many techniques on how an attacker could eavesdrop data packets flowing through the SDN network. Examples of such techniques include ARP spoofing [44], attacks to the topology discovery [45] or by physically connecting to the communication link, all of them launched from the data plane without needing access to the SDN controller.
- *Data Injection:* The attacker can inject false data into the network traffic, potentially manipulating control signals or disrupting system operations. Examples of an adversary's actions may include random attacks, carefully designed attacks, reply attacks, stealthy attacks or packet corruption attacks, to name a few.

We also consider the following limitations or assumptions:

- *Attacker Localization:* The attacker resides within the data plane, meaning they have access to the network traffic but not directly to the control plane (SDN controller).
- *Network Redundancy:* Our framework relies on the existence of network redundancy, a common feature in industrial networks designed for high availability and fault tolerance. We assume there are at least two communication paths between industrial devices. Ideally, these paths are fully disjoint, meaning they share no common network links or nodes. This configuration maximizes the effectiveness of our defense, as the compromise of one path does not affect another. In scenarios with overlapping paths, where paths share a common segment or node, an attacker compromising that segment could impact multiple replicated streams simultaneously. While this reduces the

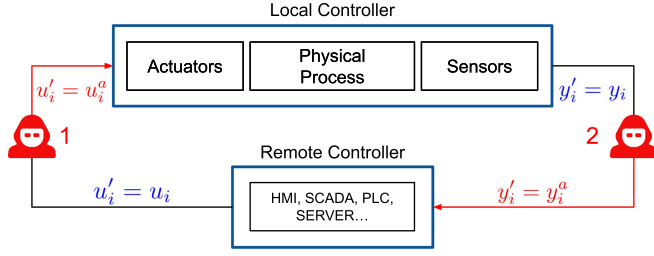


Fig. 2. Attack on control packets (1) and attack on sensor packets (2).

available path diversity, the signaling game would identify all compromised paths as suspicious. As long as at least one uncompromised path to the destination exists, it can be selected for secure data delivery.

- *Partial Network Compromise*: The attacker can compromise multiple network paths simultaneously, but not the entire network at once. This reflects a scenario where attackers may gain access to specific segments/paths of the network, but face challenges achieving complete control.

FDI Attack Process: We adopt a threat model consistent with established research on FDI attacks in control systems [5], [46], [47]. This model presumes an attacker has achieved a Man-in-the-Middle (MitM) position on a network communication path, enabling them to eavesdrop on, intercept, and modify or inject data packets in transit. The attacker's goal is to manipulate the physical process by sending false data to critical devices. As shown in Fig. 2, this can manifest as tampering with sensor readings sent to a controller ($y'_i(t) = y_i^a(t)$) or altering control commands sent to an actuator ($u'_i(t) = u_i^a(t)$). By injecting carefully crafted false values, the attacker seeks to mislead the system's control logic without being detected, potentially causing physical damage or operational failure. Our work directly counters this threat by making it difficult for the attacker's injected data to be the one selected for delivery to the end device. We formalize the measured variables $u'_i(t)$ and $y'_i(t)$ as

$$u'_i(t), y'_i(t) = \begin{cases} u_i(t), y_i(t) & \text{for } t \notin T_a \\ u_i^a(t), y_i^a(t) & \text{for } t \in T_a \end{cases} \quad (1)$$

such that $u_i(t)$, $y_i(t)$ and $u_i^a(t)$, $y_i^a(t)$ correspond to the i^{th} original and modified/tampered packet at $t \leq 0 \leq T$ respectively. T is the duration of the simulation, and T_a is the duration interval of the attack.

The attacker can decide whether to remain idle on the network or compromise the integrity of the industrial communication.

IV. DEFCLON: AN SDN-ENABLED PROACTIVE AND ADAPTIVE DATA REPLICATION FRAMEWORK

In this section, we introduce the SDN-based data replication framework. First, we provide an overview of the proposed defense architecture. Next, we introduce the proactive path activation and data replication mechanism, which plays a critical role in the proposed framework by distributing multiple copies of network packets over different paths. Finally, we present the

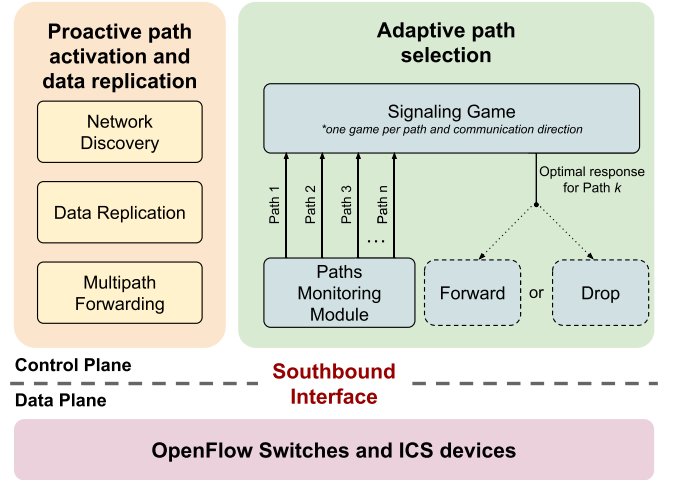


Fig. 3. DEFCLON architecture and components overview.

adaptive path selection algorithm, which uses a signaling game model to select which paths are safe to reach the destination device.

A. Architecture

In order to explain the proposed framework in detail, we provide an overview of the architecture and the main components of DEFCLON. Fig. 3 presents the overall architecture which includes two main components: the proactive path activation and data replication, and the adaptive path selection component. These two components are centrally implemented in the SDN controller. The southbound interface serves as the communication link between the SDN controller (control plane) and the forwarding devices in the network (data plane). It allows the controller to instruct the devices on handling network traffic based on the defined policies and rules. DEFCLON uses OpenFlow [48] as southbound interface communication protocol. OpenFlow enables the controller to configure flow tables within the forwarding devices, facilitating the dynamic control over packet forwarding behavior.

The proactive path activation and data replication component is responsible for managing the data replication through different paths of the network. Paths are randomized over time, creating intermittent data forwarding on the paths and making it difficult for eavesdroppers to gather complete and continuous information. This component performs the following three key functions:

- *Network Discovery*: Implements network discovery to identify hosts and links in the network. Hosts are discovered through Address Resolution Protocol (ARP, RFC826) messages and links are discovered through Link Layer Discovery Protocol (LLDP, IEEE 802.1AB) messages. The network is continuously monitored to identify changes and to maintain an updated topology view. It is important to note that while automated discovery is used for convenience, it is not a strict requirement. In environments where protocols like ARP are restricted, the network topology

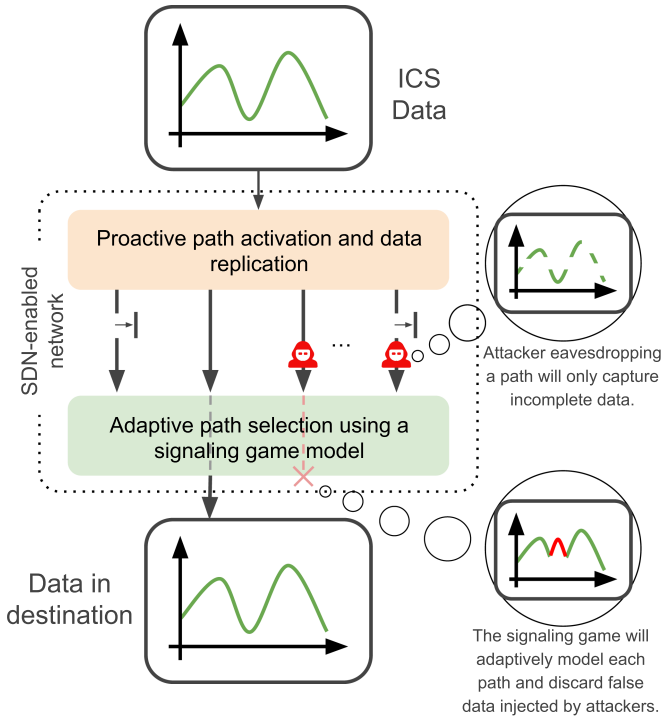


Fig. 4. Proactive and adaptive components of DEFCLON. The proactive component consists of replicating data from a subset of network paths that are randomized over time. The adaptive component consists of modeling the behavior of each path with a signaling game model to dynamically forward or drop the paths.

graph can be manually configured and provided to the SDN controller. DEFCLON's core defense mechanisms only require an updated network graph and are independent of the discovery method used.

- **Data Replication:** Selects a group of paths, performs copies of the original ICS data and sends each copy from selected paths. This module is also the responsible for changing the path selection over time.
- **Multipath forwarding:** Manages forwarding rules in the network that send the replicated data from the source to the intended destination from the set of paths selected by the *Data Replication* module.

The adaptive path selection component is the responsible for monitoring the behavior of each path using the signaling game model and selecting a single path to send data to the destination end-device. To achieve this, this component assigns a status to each path: *forward* or *drop* status. On the one hand, paths in *forward* state are eligible to be selected to send data to the destination end-device. From the set of paths in *forward* state, a single path is randomly selected to reach the destination device. On the other hand, paths in *drop* state are not eligible to reach the destination end-device. The status of each path is updated in real-time using a signaling game model.

Fig. 4 provides a visual representation of the different layers of defense in DEFCLON. An eavesdropper that intercepts network packets in a path gathers intermittent data caused by the proactive path activation and data replication. Meanwhile, an attacker attempting FDI attacks on a path will face significant obstacles in

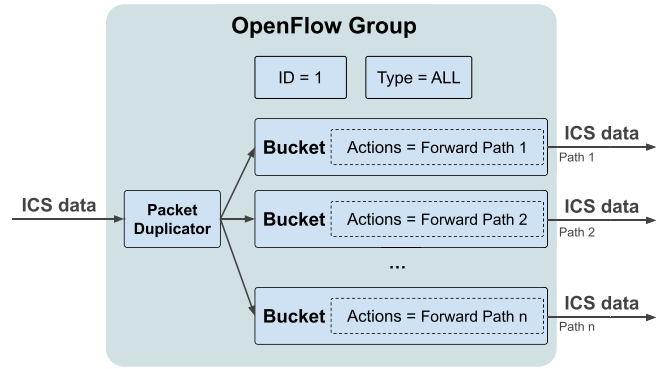


Fig. 5. ICS data replication with an ALL type OpenFlow group.

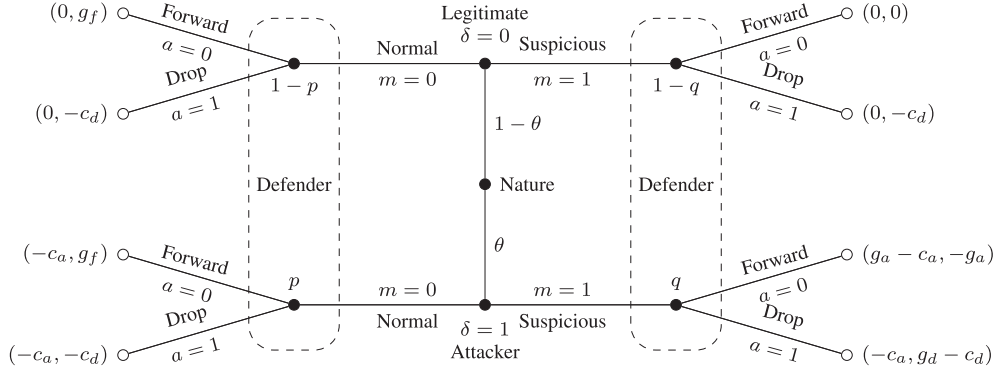
reaching the target device, thanks to our adaptive path selection mechanism. More details on these two components are provided in Sections IV-B and IV-C.

B. Proactive Path Activation and Data Replication

DEFCLON utilizes a multi-layered defense strategy. It begins with an initial layer that randomly replicates industrial network traffic across disjointed communication paths within the network. During a predetermined time interval, the SDN controller selects a random subset of available paths from the network topology to replicate data from the source to the destination device. Once the interval ends, a different and randomly selected group of paths takes over the data transmission. By adopting this approach, if an attacker manages to compromise one of the paths, their access to network traffic becomes intermittent, restricting their ability to acquire new and continuous information.

Data replication is implemented using a utility called OpenFlow *groups*. OpenFlow *groups* are a utility within the OpenFlow protocol (available since version 1.1) designed to facilitate data replication and distribution in an SDN environment. These groups are particularly useful for tasks such as multicast and data replication. By specifying a set of actions within a group (also known as *buckets*), administrators can ensure that certain packets are replicated and forwarded to multiple destinations simultaneously. This is crucial for applications where data needs to be distributed to multiple recipients or where redundancy is necessary. As defined by the OpenFlow protocol specification, there are many types of OpenFlow *groups*. In our particular case, we use the “ALL group”, which takes any received packet as input and duplicates the packet to be operated independently by each group of actions (*buckets*). The buckets within the group are configured to forward data packets to its corresponding path. The internals of an OpenFlow *group* (ALL type) is presented in Fig. 5.

The OpenFlow *group* responsible for data replication operates within the initial forwarding device, where it intercepts all data transmitted by the source ICS device. Subsequently, this group distributes multiple copies of the data packets along the designated paths established within the group. Additionally, we include an extra *bucket* in the group to mirror each packet to

Fig. 6. Extensive form representation of the signaling game $\mathcal{G}$.

the SDN controller. This allows the controller to store historical communication packets, which the signaling game model can later utilize to identify suspicious activities within each path. This technique, commonly known as port mirroring or Switched Port Analyzer (SPAN), is a common practice in industrial environments for performing packet monitoring or analysis (NISTIR 8219 [49]).

This data replication strategy is particularly synergistic with ICS design principles. Industrial networks commonly feature path redundancy to ensure fault tolerance and high availability. DEFCLON leverages this existing architectural feature for security. By proactively replicating traffic across these redundant paths, we introduce uncertainty for an adversary without altering the fundamental network topology or the high-availability requirements of the control process.

C. Adaptive Path Selection

The second layer of defense in DEFCLON is the adaptive path selection, which ensures that only a single, non-compromised path delivers data to the destination device. This is achieved in two steps. First, a filtering mechanism determines which paths are suitable for data forwarding. It uses a signaling game model to continuously monitor and assign a status of *forward* or *drop* to each active path. Second, from the filtered set of paths currently in the *forward* state, a single path is selected randomly to transmit its packet to the destination. This ensures that even among trusted paths, the final selection remains unpredictable.

As data flowing from different network paths generate uncertainty, and we have incomplete information if a specific path is operated by an attacker or not, we propose a signaling game model to help select the optimal status for each path (*drop* or *forward*). The model is an attacker-led multi-stage signaling game in which the defender monitors each path individually and updates the status of each path every time a packet is forwarded. At the last exit node, where the different paths converge and a single path is selected to send data to the destination device, the node mirrors the packets received from each path to the SDN controller using the OpenFlow protocol, which triggers the execution of the signaling game in each path, updating their status. The SDN controller maintains one signaling game per path and communication direction, updating

TABLE II
NOTATION TABLE

Notation	Description
$\mathcal{G}$	Signaling game
S, R	Sender, Receiver
$m \in M$	Messages $\{0, 1\}$
$a \in A$	Actions $\{0, 1\}$
N	Total number of available paths $N = N_f + N_d$
N_f	Number of paths in forward state
N_d	Number of paths in drop state
$\psi(r, r_n)$	Replicated data dissimilarity $\{0, 1\}$
$\delta \in \Delta$	Sender types $\{0, 1\}$
$\theta_n(t)$	Belief of n -th path at step t
$\mu^P(\delta, m, a)$	Payoff function for player P where $P \in \{S, R\}$
$\varepsilon^S(m \delta)$	Sender strategy m given its type δ
$\varepsilon^R(a m)$	Receiver strategy a given m
g_f	Gain of R for taking $a = 0$
g_d, c_d	Gain and cost of R for taking $a = 1$
g_a, c_a	Gain and cost of S for sending $m \in M$

the status each time the node mirrors a packet from a path to the controller.

The adaptive path selection component randomly selects a single path from the set of paths in *forward* state to send data to the destination device, and discards paths in *drop* state from the selection pool. Each path is continuously and individually monitored by a signaling game model, which adaptively updates the path's state (*drop* or *forward*) in real-time.

In the following, we first provide an overview of the signaling game model by defining players, messages, and action sets. Next, we define the belief model and the payoff functions of both players. Finally, we analyze the game processes between players, and explore optimal strategies by calculating the Perfect Bayesian Nash Equilibria (PBNE) [50] in the game.

1) *Game Overview*: The proposed signaling game model, represented in Fig. 6, constitutes a non-cooperative game where each path is considered as an individual sender that could be a legitimate or an attacker. An attacker, who has gained access to a communication path, could tamper industrial network packets to misguide the control algorithm or inject corrupted network packets to generate errors in the communication. The notations used in this paper are summarized in Table II, and the necessary definitions in this game model are listed as follows.

Players: The interaction between the sender (S) and receiver (R) constitutes a non-cooperative game, where the attacker

assumes the role of the sender, and the defender receives the messages in the form of replicated packets. The defender applies MTD by replicating traffic through different paths in the network and selecting a single path to reach the destination device. Due to the inability to distinguish between a legitimate user and an attacker, the defender must rely on the information in the paths to decide. The signaling game $\mathcal{G}$ is played individually for each path regardless of the sender type $\delta \in \Delta = \{0, 1\}$, where $\delta = 0$ is considered a legitimate user whereas $\delta = 1$ represents an attacker.

Messages: Signaling game $\mathcal{G}$ is considered a multistage game where the sender sends a set of possible messages $m \in M = \{0, 1\}$ and the defender receives these signals to take the corresponding action. Various messages can be formed from different configurations, and they have the ability to influence the defender's viewpoint on whether a path is under attack or not. Since the space of possible messages is large, the game focuses on two specific messages: whether the actual replicated packet differs from the originally forwarded packet or not. Based on this, message $m = 1$ indicates that the path is suspicious, whereas $m = 0$ represents that the path is under normal operation.

For example, a player $\delta = 1$ (attacker) may choose to remain idle on the network without modifying packets ($m = 0$), or to attack by modifying or corrupting packets in transit ($m = 1$). Similarly, a path under normal operation $\delta = 0$ (legitimate) may transmit packets correctly ($m = 0$), or packets may be accidentally corrupted ($m = 1$) in transit due to electromagnetic interferences, physical medium issues, signal attenuation, noise, congestion, jitter, or hardware/software malfunctions, to name a few.

Actions: After receiving a message $m \in M$ from the sender, the defender selects an action for the path by choosing $a \in A = \{0, 1\}$. Action $a = 0$ denotes that the path can be selected to be forwarded to the destination, while $a = 1$ signifies that the path must be dropped.

2) Belief Model: In signaling game $\mathcal{G}$, the defender cannot determine whether the traffic comes from a legitimate user or an attacker. Consequently, the defender monitors the signals from each path and uses that information to form a belief function (θ) concerning the sender's identity on each observation. The defender has a belief per path, i.e., if the traffic of a communication is replicated from five paths, the defender will have five different beliefs for that communication, one per path. The belief values range between 0 and 1, 0 representing that no sign of suspicious activity has been detected. The belief function comprises two parts: the initial belief that is calculated statically at the beginning of the game at $t = 0$ and the dynamic belief function used to update the belief each time the defender receives a mirrored packet from a path at $t > 0$.

Initial belief: At the beginning, there is no information about any of the paths, and the same level of suspicion is applied to all of them. Thus, at step $t = 0$, the defender assumes that each path is legitimate and free from any form of attack. The initial belief for each path at $t = 0$ is initialized and calculated as follows:

$$\theta_n(0) = 0, n \in N. \quad (2)$$

This belief initialization can be adjusted if the defender has knowledge of whether some communication paths are more vulnerable to attack, by setting a higher initial belief if this is the case.

Dynamic belief: The dynamic belief of a given path is crucial for evaluating the expected payoffs for both players. This dynamic belief, denoted by $\theta_n(t)$, is recalculated at each step (i.e., $t > 0$) as the game is played. The (3) describes how a player's dynamic belief is computed, which involves two weighted factors that are combined to produce a score. Since the weights are normalized values, the sum of the two weights is $w_1 + w_2 = 1$ (i.e., 100%).

$$\theta_n(t) = \theta_n(t-1) * w_1 + \psi(r, r_n) * w_2, n \in N, t > 0. \quad (3)$$

The first factor represents a path's prior belief $\theta_n(t-1)$, weighted by w_1 . When evaluating a client's behavior, it is important to consider their history of suspicious or legitimate actions. This is because a single deviation from their typical behavior cannot completely negate the previously observed and learned patterns. In other words, the client's current action cannot completely override the belief in their historical behavior. Therefore, to evaluate their overall performance, the prior belief is essential to take into account the sender's previously observed behavior.

The second factor of the dynamic belief function measures the dissimilarity between a given packet r_n and the original packet r previously mirrored from the initial node and stored in the SDN controller. In the context of ICS, the dissimilarity metric, denoted by $\psi(r, r_n)$, is especially powerful. Unlike IT networks, which have highly variable traffic, ICS communications are often deterministic and cyclical (e.g., periodic sensor polling). Therefore, a deviation captured by this metric is a strong indicator of manipulation. The dissimilarity metric is binary and is weighted by a parameter w_2 , where $\psi(r, r_n)$ is set to 0 if the packets are identical and 1 otherwise.

3) Payoff Model: In the following, we define the payoff model considering all possible strategy profiles and different player types. Let $\mu^P(\delta, m, a)$ represent the payoff function of player P , where $P \in \{S, R\}$. Both players aim to maximize their payoffs, maintaining a balance between gain and cost.

Receiver: If the defender chooses the drop strategy (i.e., $a = 1$) in response to a suspicious packet sent by an attacker (i.e., $\delta = 1$), the defender effectively defends the target system. Let's consider N as the total number of available paths, N_f as the number of paths in the forward state, and N_d as the number of paths in the drop state.

The receiver's numerical gain for defending against an attacker who plays suspiciously can be quantified using the parameter $g_d = 1 - \frac{N_d}{N}$. A higher value of N_d results in a lower gain from dropping, while a lower value of N_d results in a higher gain. Dropping a path also has a cost to the defender, as it reduces the size of the path selection pool. This cost is denoted by the parameter $c_d = \frac{N_d}{N}$, which represents the ratio of paths in the drop state to the total number of available paths, N . A higher value of N_d increases the cost of dropping another path, while a lower value of N_d decreases the cost. Therefore, whenever the defender decides to drop a path, it will incur a

cost of c_d . Critically, the cost of dropping a path (c_d) lies in the reduction of path diversity, which may weaken the industrial network's resilience and reliability, particularly in real-time control operations.

It is important to note that the defender only gets the gain for dropping a path (g_d) if the player type is $\delta = 1$ and the signal is $m = 1$. This gain from correctly dropping a compromised path (g_d) translates directly to the prevention of physical process disruption or equipment damage.

The defender's numerical gain for selecting forward in a path when the received signal is $m = 0$ can be expressed as the parameter $g_f = 1 - \frac{N_f}{N}$. This gain is influenced by the value of N_f , where a higher value of N_f corresponds to a lower gain, while a lower value of N_f corresponds to a higher gain. Functionally, the gain from forwarding a legitimate packet (g_f) represents maintaining the operational integrity and stability of the physical process. The defender will only gain g_f when players $\delta = 0$ and $\delta = 1$ send signal $m = 0$. If a legitimate sender $\delta = 0$ plays suspicious $m = 1$ (i.e., packet corruption during transmission) and the defender plays $a = 0$, the defender action will not result in any gain, as it may generate a delay in the network due to the retransmission carried out by the transport layer protocol (i.e., if TCP is used).

Sender: The attacker's main goal is to deliver the spoofed/tampered packets to the target device. The attacker's gain (g_a) is achieved only if their malicious packet is selected to reach the end-device, allowing them to directly influence the physical process. Because of this, the attacker gets a significant gain g_a by sending a signal $m = 1$, and the defender decides to apply the action $a = 0$. However, being active on the network costs the attacker c_a . Thus, the attacker incurs a cost of c_a every time a signal is sent, and gains g_a only when the signal is $m = 1$ and the defender chooses $a = 0$. This tradeoff between gain and cost characterizes the attacker's strategy in the network, while our defense aims to make this payoff difficult to achieve.

4) Signaling Game Analysis: In games with complete information or non-Bayesian games, a strategy profile is said to be a Nash Equilibrium (NE) if every strategy in that profile is the best response to every other strategy. In such games, each player knows the rules of the game and has complete information about the other players, strategies, and payoffs. However, in Bayesian games, players have incomplete information and uncertainty about the other players types, which affects their expected payoffs. In these games, players aim to maximize their expected payoffs given their beliefs about the other players. Therefore, a strategy profile in a Bayesian game is a PBNE if every players' strategy is the best response to their beliefs about the other players strategies and types, and the beliefs are consistent with Bayes rule. PBNE is a refinement of NE that takes into account players incomplete information and the resulting uncertainty in their decision-making. The conditions for a PBNE are the following:

- *Condition 1:* After observing a message m from the sender, the receiver must have a belief $\theta(\delta|m)$ about the sender that satisfies

$$\sum_{\delta \in \Delta} \theta(\delta|m) = 1 \quad \text{and} \quad \theta(\delta|m) \geq 0. \quad (4)$$

- *Condition 2:* For all message types $m \in M$, receiver's strategy ε^R must maximize his expected payoff given the belief $\theta(\delta|m)$ about which sender type sends m , such that

$$\varepsilon^R \in \max_{m \in M} \sum_{\delta \in \Delta} \theta(\delta|m) \mu^R(\delta, m, a). \quad (5)$$

- *Condition 3:* The sender's (legitimate or attacker) strategy ε^S must maximize his payoff given the defender's action $a \in A$, such that

$$\varepsilon^S \in \max_{m \in M} \mu^S(\delta, m, a). \quad (6)$$

- *Condition 4:* If there is a type δ that satisfies senders strategy $m^* \in M$, the defender's belief about the sender type corresponding to message m must follow the Bayes rule so that

$$\theta(\delta|m) = \frac{p(\delta)}{\sum_{\delta \in \Delta^*} p(\delta)}. \quad (7)$$

where Δ^* refers to the set of senders that sent m^* and $p(\delta)$ indicates the prior probability of sender being type δ .

Definition 1: A PBNE in a signaling game is a pair of strategies $\varepsilon^S(m|\delta)$ and $\varepsilon^R(a|m)$, and a belief $\theta_n(t)$ that satisfies conditions 1-4.

In a signaling game, there can be multiple PBNE, which are sets of stable strategies against deviations by either player. These PBNEs can be divided into two categories: separating equilibrium and pooling equilibrium. In a pooling equilibrium, the sender uses the same signal for multiple types $\varepsilon^S(m|0) = \varepsilon^S(m|1)$, and the receiver chooses the same action regardless of the signal received. In a separating equilibrium, on the other hand, the sender uses a particular signal to convey information about their type to the receiver, and the receiver uses this information to choose the appropriate action. In a separating equilibrium, the sender's signal is different for each type $\varepsilon^S(m|0) \neq \varepsilon^S(m|1)$.

In order to identify the existing equilibrium profiles in the signaling game $\mathcal{G}$, the conditions that lead to pooling or separating equilibriums are examined next. The values of the payoffs may vary depending on the network topology or number of paths available in a communication between two devices, requiring to determine the existence of different equilibrium profiles in the game.

Theorem 1: There is a pooling equilibrium $\forall \theta$ on $m = 1$ in the signaling game $\mathcal{G}$.

Proof: Suppose both sender types follow the same strategy $m = 1$. In this scenario, the defender's belief about the sender type is determined following Bayes rule. The defender's expected payoff μ^{R*} for playing $a = 0$ based on its belief θ is represented by the following equation:

$$\mu_{a=0}^{R*} = \theta(-g_a) + (1 - \theta)(0). \quad (8)$$

Similarly, the following equation represents defender's expected payoff μ^{R*} for playing $a = 1$:

$$\mu_{a=1}^{R*} = \theta(g_d - c_d) + (1 - \theta)(-c_d). \quad (9)$$

By comparing the expected payoffs of the defender for each of the action sets mentioned above, a belief threshold, denoted as

θ^* , can be formulated by performing $\mu_{a=0}^{R*} = \mu_{a=1}^{R*}$ calculation, obtaining the following result:

$$\theta^* = \frac{c_d}{g_d + g_a}. \quad (10)$$

Assuming that $\theta^* \geq 0$, the following two cases represent a dominant strategy for the defender in this pooling equilibrium:

- $\theta \geq \theta^* = \frac{c_d}{g_d + g_a}$. Action $a = 1$ dominates $a = 0$ when obtaining a signal $m = 1$ and the condition $\theta \geq \theta^*$ is satisfied. This can be verified by comparing expected payoffs defined in (8) and (9). Taking into account the possible deviations of the sender, it can be observed that both sender types have no incentives to deviate. The attacker and the legitimate user will obtain the same payoffs in both message types, $-c_a$ and 0 respectively.
- $\theta \leq \theta^* = \frac{c_d}{g_d + g_a}$. Action $a = 0$ dominates $a = 1$ when obtaining a signal $m = 1$ and the condition $\theta \leq \theta^*$ is satisfied. Similar to the previous case, incentives of the senders for deviating from $m = 1$ must be checked when the defender plays $a = 0$ in the pooling equilibrium. As observed in the payoffs, neither type of sender has any incentive to deviate from its strategy. In the case of the attacker, the payoff will drop from $g_a - c_a$ to $-c_a$, while the payoff of a legitimate user will remain 0.

Theorem 2: There is no pooling equilibrium $\forall \theta$ at $m = 0$ in the signaling game $\mathcal{G}$.

Proof: Following a pooling strategy in message $m = 0$, defender's best response is $a = 0$, obtaining a payoff of g_f with both sender types. Hence, deviations from the sender should be taken into account considering receivers optimal response in this pooling strategy. As a legitimate user (i.e., sender type $\delta = 0$) obtains 0 in all strategy profiles, let only consider attackers $\delta = 1$ possible deviations from the equilibrium. If it deviates from $m = 0$ to $m = 1$, the attacker will increase its payoff from $-c_a$ to $g_a - c_a$ when the defender plays $a = 0$. Because of this, there is no pooling equilibrium at $\forall \theta$ on $m = 0$.

Theorem 3: There are two separating equilibriums on $\{(\delta = 0, m = 0, a = 0), (\delta = 1, m = 1, a = 1)\}$ and $\{(\delta = 0, m = 0, a = 0), (\delta = 1, m = 1, a = 0)\}$ in the signaling game $\mathcal{G}$.

Proof: Let's suppose a separating strategy where player type $\delta = 0$ sends $m = 0$ and player type $\delta = 1$ sends $m = 1$. Best response of the defender for sender type $\delta = 1$ sending $m = 1$ can be obtained by comparing $g_d - c_d$ and $-g_a$ payoffs, resulting in:

$$g_d^* = c_d - g_a. \quad (11)$$

Payoffs comparison leads to the following two scenarios:

- $g_d \geq g_d^* = c_d - g_a$. The best response for the defender is $a = 1$ when player $\delta = 1$ plays $m = 1$ and $a = 0$ when player $\delta = 0$ plays $m = 0$. Checking possible sender strategy deviations, both $\delta = 0$ and $\delta = 1$ will not have incentives to deviate as they will obtain the same payoff, 0 and $-c_a$ respectively.
- $g_d \leq g_d^* = c_d - g_a$. In this scenario, the best response for the defender, when player $\delta = 1$ plays $m = 1$ and player $\delta = 0$ plays $m = 0$, is $a = 0$. Same as the previous

Algorithm 1: Adaptive Path Status Selection Algorithm.

Input: Game parameters: $w_1, w_2, g_f, g_d, c_d, g_a, c_a$
 Path: n

Output: Optimal response to path:
 $R_n(t) \in \{\text{forward}, \text{drop}\}$

```

1 Initialize belief  $\theta_n(0) = 0$ 
2 for step  $t = 1$  to  $T$  do
3   Compute  $g_f, g_d, c_d, g_a, c_a$ 
4   Compute  $\theta_n(t)$ 
5   Compute  $\theta^*$  (Theorem 1)
6   Compute  $g_d^*$  (Theorem 3)
7   if  $m = 1$  and  $(\theta_n(t) \geq \theta^* \text{ or } g_d \geq g_d^*)$  then
8      $R_n(t) \leftarrow \text{drop}$ 
9   else
10     $R_n(t) \leftarrow \text{forward}$ 
11 Return  $R_n(t)$ 

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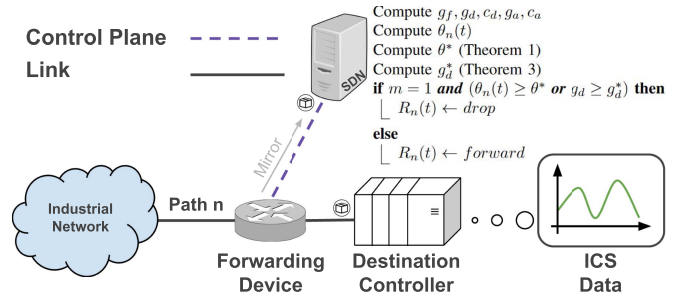


Fig. 7. The adaptive path status decision-making process in DEFCLON.

scenario, senders have no incentive to deviate from their strategy.

Theorem 4: There is no separating equilibrium on $\{(\delta = 0, m = 1), (\delta = 1, m = 0)\}$ in the signaling game $\mathcal{G}$.

Proof: Consider a separating strategy where player $\delta = 0$ plays $m = 1$ and player $\delta = 1$ plays $m = 0$. In this case, the best response of the defender to a sender $\delta = 1$ playing $m = 0$ and a player type $\delta = 0$ playing $m = 1$ is $a = 0$. Focusing on sender type $\delta = 1$ possible deviations, we can conclude that there is not a separating equilibrium in this scenario. If the sender type $\delta = 1$ deviates by playing $m = 1$ instead of $m = 0$ and the defender responds to that message with $a = 0$, the payoff of the sender type $\delta = 1$ will be increased from $-c_a$ to $g_a - c_a$. Because of this, there is no separating equilibrium on $\{(\delta = 0, m = 1), (\delta = 1, m = 0)\}$.

Theorems 1 to 4 represent all possible PBNE in the proposed signaling game $\mathcal{G}$. The process of adaptively monitoring and updating the status of the paths is represented in Algorithm 1 and Fig. 7, which details the decision logic and process for assigning a forward or drop status to an individual path.

V. EXPERIMENTAL SETUP

In this section, we present the experimental setup for validating DEFCLON: the physical process, the network topologies, and the FDI attack implementation process.

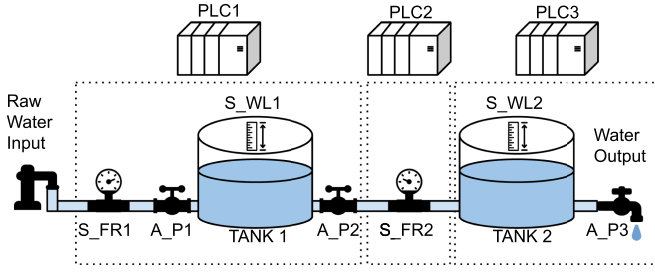


Fig. 8. First stage of the Secure Water Treatment (SWaT) testbed implementation in MiniCPS.

TABLE III
SENSORS AND ACTUATORS FUNCTION, MEASUREMENTS AND STATUS

Sensor/Actuator	Function	Measurement/Status
S_FR1	Flow rate measurement	m^3/h
S_FR2	Flow rate measurement	m^3/h
A_P1	Water flow control	open (1), close (0)
A_P2	Water flow control	open (1), close (0)
A_P3	Water flow control	open (1), close (0)
S_WL1	Water level measurement	millimeters
S_WL2	Water level measurement	millimeters

TABLE IV
WATER TANKS LEVELS CONTROL LOGIC THRESHOLDS

Threshold	Description	Value (millimeters)
HH	Overflow threshold	1200mm
H	High threshold	800mm
L	Low threshold	500mm

A. Physical Process

The experimental evaluation is conducted within MiniCPS¹ framework, a platform for cyber-physical system simulation and testing introduced by Antonioli et al. in their work [51]. This framework includes the simulation of the first of the six stages of the Secure Water Treatment (SWaT) system [52], which is widely used for research and experimentation in the area of ICS security.

The first stage of the SWaT testbed, illustrated in Fig. 8, involves two water tanks, TANK 1 and TANK 2, two flow rate sensors (S_FR1, S_FR2), two water level sensors (S_WL1, S_WL2), and three actuators (A_P1, A_P2, A_P3). These actuators are responsible for regulating the water flow, ensuring it progresses to the subsequent stages of the treatment process. Table III provides a detailed breakdown of the sensors and actuators, including their functions, measurements, and status-related information.

The control process depends on predefined water level thresholds, which are listed in Table IV. The primary objective of the control logic is to maintain the water level within the specified high (H) and low (L) thresholds in both TANK 1 and TANK 2, ensuring that it remains within an optimal range. Furthermore, there is an overflow threshold, denoted as HH, which represent the upper maximum limit of the water level.

PLC1 is tasked with monitoring the water level data received from sensor S_WL1 and comparing it against these thresholds. Based on this comparison, it determines whether to open or close actuator A_P1. Additionally, PLC1 communicates with PLC2 to obtain sensor data from S_FR2 and with PLC3 to acquire readings from sensor S_WL2. By comparing these sensor readings against predefined thresholds, PLC1 determines the appropriate state for actuator A_P2 (open or close).

PLC3 is responsible for managing the water level data from sensor S_WL2, which measures the water level in TANK 2. The primary goal here is to prevent tank overflow by comparing the sensor reading with the HH threshold. In the event that the water level exceeds the maximum capacity of 1200 mm (HH), PLC3 opens actuator A_P3 to reduce the risk of overflow and tries to reduce the water level until it reaches the H threshold.

The simulation starts with an initial water level of 800 mm in TANK 1 and 400 mm in TANK 2. Since the water level in TANK 2 is below the threshold L (500 mm), the control process starts with the stabilization phase by opening actuator A_P2, and sending water from TANK 1 to TANK 2 in order to reach the water level indicated by threshold H (800 mm). The system is considered stabilized when both water tanks reach a water level of 800 mm (indicated by the H threshold). When the water level in TANK 1 falls below the L threshold (500 mm), PLC1 opens actuator A_P1 until the water level reaches the H threshold (800 mm).

The PLCs communicate using the Ethernet/IP protocol, which facilitates the exchange of information to control and manage the water treatment process.

B. Industrial Control Network Topology

We conducted the experiments in mesh and ring topologies, each consisting of 6 nodes (Fig. 9). These network architectures are widely used and readily available for industrial applications, as evidenced by technical documentation and reports from various industrial equipment vendors [53], [54], [55].

A mesh topology (Fig. 9(a)) is a network design where each device in the network is interconnected to every other device, creating a redundant and highly fault-tolerant system. Within the mesh network, each device serves as a node, and it communicates directly with all other devices in the network. This redundancy allows for data to flow through multiple paths, ensuring that even if one link or device fails, communication can continue through alternative routes. The mesh topology is suitable for applications such as process control, where real-time data exchange is essential.

In contrast to the mesh topology, a ring topology (Fig. 9(b)) involves connecting devices in a circular arrangement, where each device is connected to precisely two others, forming a closed loop. One advantage of the ring topology is that it is relatively simple to set up and manage compared to the mesh topology, as there are fewer connections to configure. However, it is less fault-tolerant; if one device or connection in the ring fails, it can disrupt the entire network.

Furthermore, it is worth noting that in one of the experiments, we extended the number of nodes beyond the default 6 to

¹ Available at <https://github.com/scy-phy/minicps>

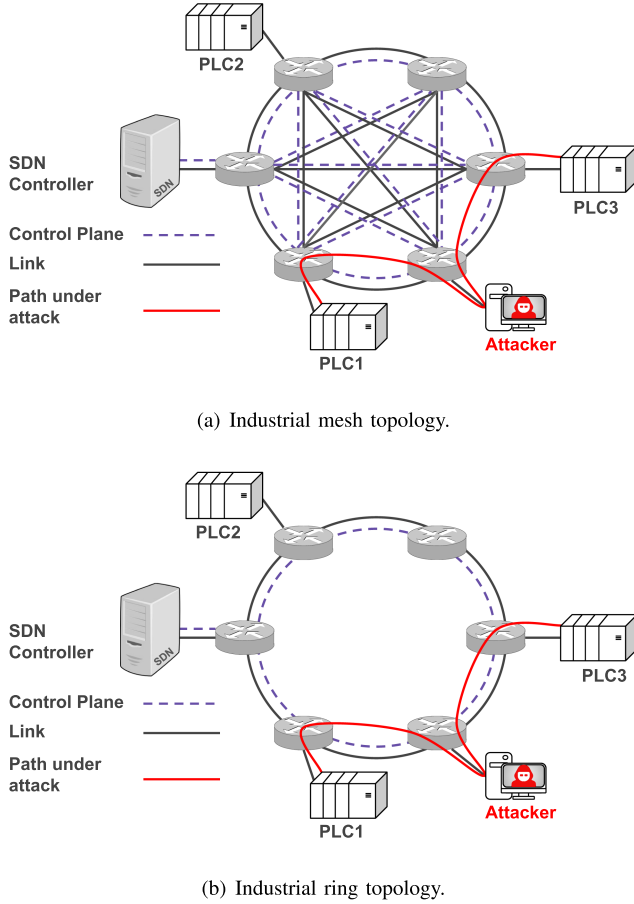


Fig. 9. Industrial control network topologies used in the experimentation: A mesh network topology (a) and a ring network topology (b).

introduce scalability and variation into the assessment. This allowed us to observe how the implementation of DEFCLON affects network performance, providing valuable insights into its suitability for different industrial network sizes.

On top of these two network topologies, we implemented DEFCLON on a single SDN controller powered by Ryu.² To ensure efficient data transmission throughout the network, we constructed the network topology using Open vSwitch³ switches. The entire network, including the SDN controller, was deployed on a single server powered by Ubuntu 22.04, which featured an Intel Xeon E5-2630 processor and 32 GB of RAM.

We define the quantitative measures for the parameters of the signaling game $\mathcal{G}$ for the path status decision-making as follows. To simplify the analysis, we stabilize the attacker's numerical gain and loss, g_a and c_a , respectively, as a normalized value of 1. We dynamically compute the remaining game parameters (g_f, g_d, c_d) as described in Section IV-C3 each time the game is played. Additionally, we set the weights in the belief model for running the experiments to $w_1 = 0.9$ and $w_2 = 0.1$. With these values, we achieve a good trade-off between past interactions and current information sensitivity, as shown in our experiments

in the next section. In one of the experiments, we test different weight values in the belief model and analyze the evolution under different configurations.

C. FDI Attack Implementation

Following the threat model described in Section III, we designed the attack with the goal of degrading performance, misleading the control algorithm, and causing damage to physical industrial equipment. The manipulated variable corresponds to the sensor S_WL2 , which measures the water level in TANK 2. In this attack, we assume that the attacker gains access to a communication path between PLC1 and PLC3 and tampers S_WL2 sensor reading data requested by PLC1 to PLC3, changing the water level value to 0.4 meters, below the optimal level defined by the threshold L setting. In response, PLC1 opens actuator A_P2 , sending water from TANK 1 to TANK 2 and eventually increasing the water level in TANK 2 until it reaches the overflow threshold.

VI. EXPERIMENTAL RESULTS

In this section, we first investigate the proactive path activation and adaptive path selection probabilities of DEFCLON in Sections VI-A and VI-B. We then test DEFCLON on the SWaT testbed in Section VI-C, and compare it to the closest works in the literature to highlight its ability to mitigate FDI attacks in industrial environments. In Section VI-D, we study the behavior of the belief model with different configurations and demonstrate the ability of the model to adapt to different scenarios. Finally, in Section VI-E we examine the end-to-end performance impact of DEFCLON in networks with different size.

A. Estimating Proactive Path Activation Probabilities

The proactive path activation aims to create uncertainty for attackers by preventing them from consistently accessing network traffic and gathering information to construct effective attacks. Considering the scenario where an attacker breaches a single communication path, the probability of network traffic being transmitted through the attacker's path is calculated using combinatorics. The binomial coefficient, denoted as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, represents the number of ways to choose k elements from a set of n elements, regardless of their order. In DEFCLON, given the availability of N paths to transmit data, the total number of combinations without repetition C_t can be computed as

$$C_t(N) = \sum_{k=1}^N \binom{N}{k}. \quad (12)$$

Next, we calculate the number of combinations for a single path (C_p) as

$$C_p(N) = \sum_{k=1}^N \binom{N-1}{k-1}. \quad (13)$$

Assuming that all paths have the same probability, the probability associated with activating a particular path to transmit

²Available at <https://github.com/faucetsdn/ryu>

³Available at <https://www.openvswitch.org/>

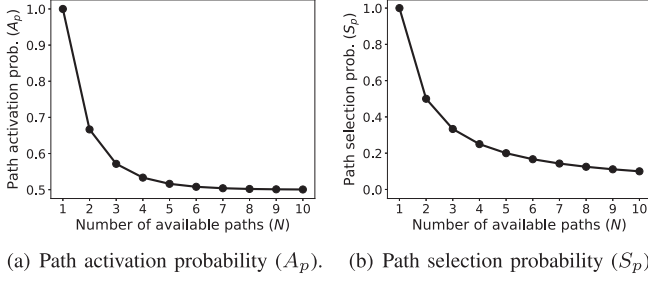


Fig. 10. Path activation (A_p) and path selection (S_p) probabilities under different number of available paths.

data, denoted by A_p , is expressed by the following equation:

$$A_p(N) = \frac{C_p(N)}{C_t(N)}. \quad (14)$$

Information-gathering is limited by DEFCLON: A random subset of paths to transmit data are periodically triggered by DEFCLON, creating intermittent data transfers. This dynamic behavior thwarts attackers with access to these paths, limiting their ability to collect information. As the number of available paths grows towards infinity ($N \rightarrow \infty$), the probability of a path being activated to transmit data approaches 50%. Consequently, there is a 50% chance on $N \rightarrow \infty$ that a path will not transmit network traffic at any given time. We illustrate this trend in Fig. 10(a).

B. Estimating Adaptive Path Selection Probabilities

Given a pool of N available paths, all of which are in a forward state, we can determine the probability of selecting a particular path to forward data to the destination device using the following calculation. Assuming an equal probability of selecting any single path for this task, the selection probability (S_p) can be expressed as

$$S_p(N) = A_p(N) \frac{\sum_{k=1}^N \frac{\binom{N-1}{k-1}}{k}}{C_p(N)} = \frac{1}{N}. \quad (15)$$

DEFCLON introduces uncertainty in path selection: As the number of available paths (N) approaches infinity ($N \rightarrow \infty$), the probability of selecting a particular path to forward data to the target device converges to 0%. Consequently, as the number of paths increases, the probability of a path being selected to forward traffic to the destination device decreases. This significantly reduces the probability of adversaries to perform FDI attacks on a path selected to reach the destination device. This trend is illustrated in Fig. 10(b).

C. Mitigating FDI Attacks in the SWaT Testbed

In Sections VI-A and VI-B, we demonstrated that the proactive components of DEFCLON inherently limit an attacker's opportunities, showing that an adversary faces significant uncertainty and a low probability of success by default. In this section, we evaluate DEFCLON's effectiveness under a worst-case scenario. We simulate a scenario in which an adversary has

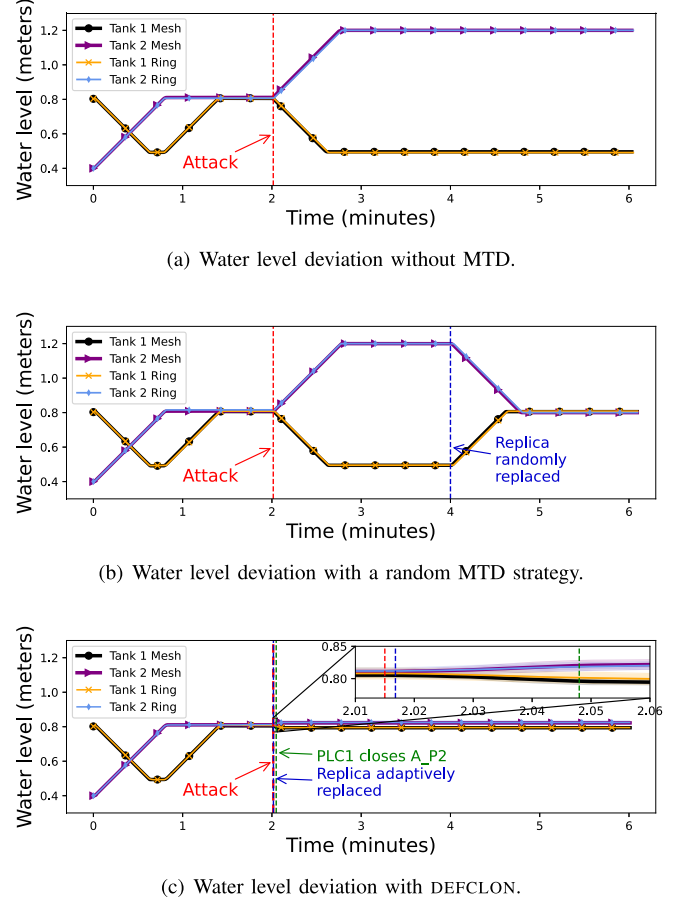


Fig. 11. Water level deviation under different network topologies and MTD implementation strategies.

a presence on a network path and initiates an attack on the path currently selected for data delivery to the end device. This approach enables us to measure the effectiveness of the signaling-game-based response in mitigating an ongoing attack.

For this, we executed the DEFCLON framework on the initial stage of the SWaT testbed, as detailed in Section V-A. Fig. 11 illustrates the temporal evolution of two water tank levels, namely TANK 1 and TANK 2, across three scenarios: (1) Water level deviation without the use of MTD or any defense mechanism. (2) Water level deviation with the random traffic replication MTD strategy proposed in [43] and [42]. (3) Water level deviation with our proactive and adaptive traffic replication MTD framework DEFCLON.

For each scenario, we conducted a total of 20 simulations, using two different network topologies: mesh and ring, as discussed in Section V-B. We performed 10 simulations for each topology type. The solid lines represent the mean water level across the 10 trials for each configuration, while the shaded area around each line in Fig. 11(c) indicates the standard deviation. The small standard deviation observed across all trials indicates a high degree of consistency and reproducibility of DEFCLON. In each simulation run, we allowed an initial 2-minute window for the control logic to stabilize the process, ensuring a water level of approximately 0.8 meters in both tanks. After

TABLE V
COMPARISON OF DEFCLON WITH OTHER DATA REPLICATION STRATEGIES AGAINST FDI ATTACKS

Comparison Aspects	MTD-IDR [42]	DMTDR [43]	DEFCLON (our proposal)
Application Domain	CPSs	Microgrids	ICSs
Network Infrastructure	Traditional Networking	Traditional Networking	SDN
Hardware Requirements	IoT Devices, Data Exchange Network	IoT Devices, Data Exchange Network	SDN infrastructure
Path Activation Strategy	Proactive/Random	Proactive/Random	Proactive/Random
Path Selection Strategy	Proactive/Random	Proactive/Random	Adaptive
Path Selection Probability (normal operation)	$S_p(N)\%$	$S_p(N)\%$	$S_p(N_f)\%$
Path Selection Probability (under attack)	$S_p(N)\%$	$S_p(N)\%$	Dropped by the signaling game = 0%

this initialization period, a simulated adversary starts with the FDI attack, following the procedure outlined in Section V-C. The moment at which the testbed is actively attacked is indicated by a vertical dashed red line. During these experiments, we assume the worst-case scenario where the attacker starts the FDI attack in the path selected to reach the destination end-device.

Examining Fig. 11(a), which illustrates the impact of the FDI attack in the absence of any defense mechanism, we observe the following sequence of events:

- At minute 2, when the attacker initiates the attack by tampering S_WL2 sensor reading packets requested to PLC3, PLC1 responds by opening the A_P2 actuator. This action results in the transfer of water from TANK 1 to TANK 2, causing an increase in the water level in TANK 2 and a corresponding decrease in TANK 1 water level.
- When the water level in TANK 2 reaches 1.2 meters, PLC3 opens A_P3 to prevent an overflow. As a result, the water level stabilizes at 1.2 meters (threshold HH), as the rate at which water leaves A_P3 and enters the tank matches.
- A similar phenomenon occurs in TANK 1. When PLC1 activates the A_P2 actuator to transfer water to TANK 2, TANK 1 water level decreases until it reaches the threshold value (L) of 0.5 meters.
- In response to this decrease, PLC1 opens the A_P1 actuator in an attempt to stabilize the water level at 0.8 meters. The reason TANK 1 water level stabilizes at 0.5 meters after the attack is due to the fact that the rate of water flowing from A_P2 matches the rate at which water enters from A_P1.

DEFCLON outperforms random MTD strategies: The comparison presented in Fig. 11 demonstrates that DEFCLON significantly outperforms both the MTD-IDR [42] and DMTDR [43] random MTD strategies. In the case of the random traffic replication MTD strategy shown in Fig. 11(b), when the attack initiates at minute 2, the water level changes in the tanks remain identical to the scenario without MTD. However, when the randomness of this strategy replaces the path under attack, at minute 4 in our simulations (indicated by a blue dashed vertical line), the control logic attempts to stabilize both tank's water levels at 0.8 meters. Conversely, when observing Fig. 11(c), it becomes evident that DEFCLON's adaptive path selection strategy, based on the signaling game model $\mathcal{G}$, is far more effective in mitigating the attack compared to the random MTD strategy. Thanks to the signaling game, the attacked path is rapidly replaced shortly after the attack begins (indicated by a vertical blue dashed line). As soon as the attacked path is replaced, PLC1 closes the actuator

A_P2 in the next cycle (indicated by a vertical green dashed line), effectively limiting the deviation in water levels compared to the random MTD strategy. Table V summarizes the main differences between the random MTD strategies (MTD-IDR and DMTDR), and our adaptive MTD proposal DEFCLON.

DEFCLON is topology-agnostic: The fact that DEFCLON performs well on different network topologies ascertains the validity of our claim about the applicability of our method to a wide range of topologies. This versatility and adaptability of DEFCLON in addressing FDI attacks on various ICS network topologies is attributed to the inherent advantages provided by SDN. SDN introduce an abstraction layer between the data plane and the control plane, allowing for enhanced flexibility and control in any network topology design. As shown in Fig. 11, DEFCLON is able to respond to FDI attacks on different ICS network topologies that had actual impact on the physical process.

D. Experimenting With the Belief Model

The belief model plays a crucial role in the decision-making process and equilibrium calculation within the signaling game $\mathcal{G}$. This study introduces a belief model composed of two factors, each weighted by w_1 and w_2 respectively (3). By fine-tuning these weights, we can observe diverse trajectories in the evolution of the belief values.

The belief model evolution is adaptable: To visualize different evolutions of the belief model, Fig. 12 represents three scenarios with different weight values. In order to capture this evolution, we conducted an attack in a path, spanning a duration of 2 minutes. The weight w_1 represents the model's memory, which gives precedence to the path's historical behavior. A higher value of w_1 makes the system more conservative. It will be slower to drop a historically reliable path based on a single suspicious packet. On the other hand, the weight w_2 represents how reactive the model is to immediate evidence. A higher w_2 value makes the system highly sensitive, causing the belief value to spike rapidly when the first anomalous packet is detected. While this allows for near-instantaneous detection, it also increases the risk of misclassifying benign packet corruption as a malicious attack. Therefore, the choice of these parameters represents a trade-off between detection speed and stability. For example, a configuration with $w_1 = 0.1$ and $w_2 = 0.9$ is highly sensitive and suitable for environments where even minor deviations are critical and network-induced errors are rare. The configuration with $w_1 = 0.9$ and $w_2 = 0.1$ is more cautious and is suitable for environments with potential network

TABLE VI
ROUND TRIP TIME (RTT) OF NETWORK PACKETS UNDER NORMAL OPERATION WITHOUT MTD AND DEFCLON OPERATION

			Normal operation without MTD				DEFCLON (our approach)			
			2 hops	5 hops	25 hops	50 hops	2 hops	5 hops	25 hops	50 hops
RTT (ms)	Ring	Avg.	6.575	17.000	64.169	118.795	6.458	17.500	65.012	119.104
		Min	6.132	13.566	56.392	110.343	6.079	13.542	56.252	111.177
		Max	7.474	21.307	71.586	138.674	9.282	22.383	71.273	139.524
		Stdv	0.148	1.599	2.156	2.502	0.243	1.646	2.154	2.601
	Mesh	Avg.	6.582	17.316	63.796	118.218	6.496	17.957	64.337	118.929
		Min	6.101	13.770	56.019	110.999	6.080	13.986	55.284	109.713
		Max	7.686	21.984	72.460	134.411	7.485	20.965	78.308	140.446
		Stdv	0.152	1.091	1.755	2.146	0.171	0.901	2.179	2.365

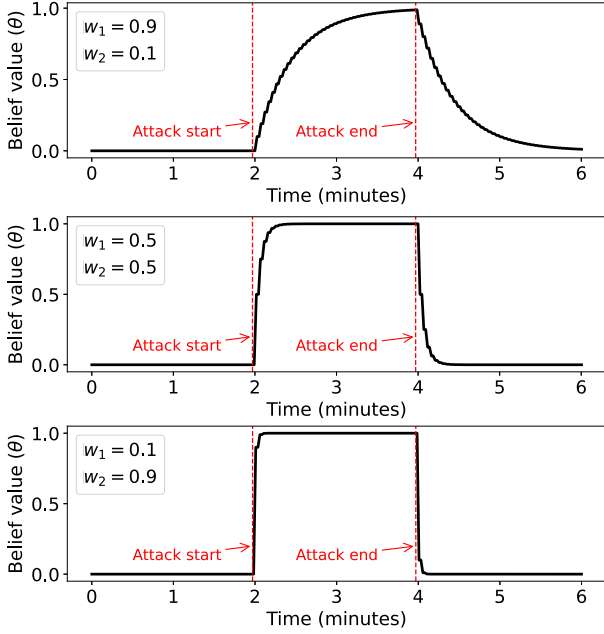


Fig. 12. Belief value evolution in a path under attack with different weight values.

noise, where the system must be confident before dropping a path.

E. Measuring Network Performance Degradation

To assess the impact of DEFCLON on network performance, we conducted a comprehensive analysis of Round Trip Time (RTT), involving the transmission of 400 ICMP traces along various network paths, each with different distances or switch hops. Our performance experiments encompassed both mesh and ring topologies. Table VI presents RTT measurements, including averages, minimums, maximums, and standard deviations. We provide these measurements for scenarios both with and without MTD, demonstrating how DEFCLON affects RTT across different path lengths and network topologies compared to a normal network operation.

DEFCLON does not degrade network performance: The comparison conducted in Table VI indicates that DEFCLON does not cause any additional performance degradation in terms of end-to-end latency compared to a normal network operation. The fact that with DEFCLON network packets are matched, processed, and forwarded directly by rules installed in the forwarding devices,

just as in a normal network operation without MTD, DEFCLON does not introduce any delay or performance degradation in the communications, regardless of the path length and the topology type.

VII. LIMITATIONS AND DISCUSSION

Controller scalability and computational load: A key consideration for DEFCLON is the computational load on the SDN controller, which handles real-time path monitoring and game-theoretic calculations. The system is designed to limit this burden inherently. First, the signaling game's calculations are computationally lightweight, involving simple arithmetic rather than resource-intensive algorithms. Second, these calculations are triggered only upon a packet's arrival at the final node before the destination, not at every hop in the network core, which significantly reduces the frequency of controller interactions. Despite this efficiency, a single, centralized controller could still become a performance bottleneck in large-scale or high-traffic industrial settings. For such scenarios, established SDN patterns offer clear scalability paths. A distributed controller architecture [56], [57] can share the monitoring and calculation load across multiple instances, enhancing performance and resilience. Additionally, incorporating a stateful data plane [58] could offload computational tasks to the network devices themselves, further reducing controller overhead.

Optimal path number selection: Refers to the challenge of determining the ideal number of paths to use in a network. The existing work assumes a fixed number of available paths without considering the dynamic nature of network conditions and requirements. This limitation can result in either underutilization of resources or insufficient redundancy. Determining the optimal path number requires a comprehensive analysis of various factors such as network size, traffic load, desired resilience level, and attack patterns. Future research in this area will focus on developing adaptive algorithms that can dynamically adjust path numbers to meet the evolving needs of modern industrial networks.

Challenges of SDN adoption in ICS environments: While SDN offers powerful capabilities for network management and security, its integration into ICS environments is still in its early stages and presents several key challenges [14]. One significant issue is real-time performance and latency; the separation of the control and data planes in SDN can introduce unacceptable delays for time-sensitive industrial processes that require millisecond-level precision. Moreover, a standard SDN

architecture with a single, centralized controller creates a single point of failure, a critical risk in ICS, which demands high availability and often necessitates a more complex, distributed, and resilient architecture to ensure reliability. The challenge is further compounded by the need to integrate SDN with legacy industrial network equipment and proprietary protocols that are not compatible with standard SDN protocols like OpenFlow, posing a major technical and logistical hurdle. Lastly, the security of the control plane is a major concern, as the centralized SDN controller becomes a high-value target for attackers, and a successful breach could give an adversary complete control over the network. Securing the controller and its communication channels is therefore a top priority, adding another layer of complexity to the overall system design.

Impact of network noise and measurement errors: A key assumption in our signaling game formulation is the complete and accurate observation of messages, specifically the packet dissimilarity metric $\psi(r, r_n)$ used to identify suspicious activity in each path. However, real-world ICS networks are subject to noise from factors like electromagnetic interference, congestion, or hardware malfunctions, which can cause benign packet corruption. Such an event could cause the system to misinterpret a legitimate path ($\delta = 0$) as sending a suspicious signal ($m = 1$). This would incorrectly increase the path's belief value ($\theta_n(t)$), potentially causing the defender to drop a healthy communication channel and incur a cost (c_d) by reducing path diversity. Our framework mitigates this through its dynamic belief model. The model's sensitivity is managed by weights assigned to historical behavior (w_1) and current packet dissimilarity (w_2). By configuring a high weight for history (e.g., $w_1 = 0.9$, as used in our experiments), the defender's belief becomes more conservative and less reactive to isolated instances of packet corruption. The selection of these weights creates an operational trade-off between the speed of attack detection and the system's stability against non-malicious network noise.

VIII. CONCLUSION

We present a multi-layered MTD defense framework that proactively and adaptively mitigates FDI attacks in the context of SDN-based ICS. Leveraging SDN central management and global view capabilities, we replicate the industrial traffic from different network paths, and in the last exit node, we select only a single path to be forwarded to the intended destination device. To comprehensively depict the interactions between attacker and defender, we propose a multi-stage signaling game to model the attack-defense process and evaluate the payoffs of both players. Furthermore, based on this game, we introduce an optimal path status selection algorithm to include or exclude paths from the selection pool for reaching the destination end-device. Our evaluation results reveal the effectiveness of the proposed framework on multiple areas: (1) It reduces adversaries' ability to intercept network traffic by up to 50%. (2) It introduces uncertainty on path selection, making difficult for adversaries to inject malicious network packets into the industrial devices. (3) The framework adaptively mitigates the impact of FDI attacks in different network topologies by rapidly replacing paths under

attack, thereby preventing malicious traffic from reaching its intended destination. (4) Our adaptable weighted belief model proves its versatility in addressing diverse scenarios and specific needs. (5) The framework operates without introducing any additional performance degradation compared to a standard network operation.

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Xabier Etxezarreta received the PhD degree, in 2024 with a thesis on software-defined networking approaches for intrusion response in industrial control systems. He is a researcher/lecturer and member with the Data Analysis and Cybersecurity research group, Mondragon Unibertsitatea. His research interests include securing critical infrastructure, specifically through networking security and the development of automated intrusion response strategies.



Federico Turrin received the PhD degree in brain, mind, and computer science from the University of Padua, in 2023, under the supervision of Prof. Mauro Conti. In 2023 he was a post doc researcher with the SPRITZ research group, University of Padua. Currently he serves as a cybersecurity engineer with SpritzMatter srl. His research interests include security of cyber-physical systems, specifically industrial control systems and vehicular systems, as well as the development of anomaly and intrusion detection. In 2022, he was a visiting researcher with SUTD in Singapore, under the guidance of Prof. Jianying Zhou and Prof. Tien Tuan Anh Dinh.



Iñaki Garitano received the MSc degree in telecommunication engineering and the PhD degree in behavioral modeling for anomaly detection in industrial control systems. He is a researcher/lecturer with the Intelligent Systems for Industrial Systems research group, and member with the Data Analysis and Cybersecurity research area of Mondragon Unibertsitatea. Before joining Mondragon Unibertsitatea in 2015, he worked with Universitetsenteret på Kjeller (UNIK) as postdoctoral researcher. In addition, he is the CTO of the Basic Internet Foundation. His main

research interests include in the area of Cybersecurity applied to industrial automation and control systems and information and communication networks including Internet of Things, and application container technologies. He currently participates in several European, national and regional level public funding projects.



Mikel Iturbe received the MSc degree in ICT security from the Open University of Catalonia, Barcelona, Spain, in 2013, and the PhD degree from Mondragon Unibertsitatea, Arrasate-Mondragón, Spain, in 2017, where he worked on data-driven intrusion detection in industrial networks. He is a lecturer and a researcher with Mondragon Unibertsitatea. He is currently part of the Data Analysis and Cybersecurity Research Group. His main research interest is related to cybersecurity, primarily in the industrial sector. The main lines he works on are industrial control system

security, embedded security, and software security. He also works in exploring novel data-driven applications for cybersecurity.



Urko Zurutuza received the PhD degree from Mondragon Unibertsitatea, in 2008, in collaboration with the Zürich IBM Research Lab. He is the principal investigator with the Intelligent Systems for Industrial Systems research group, and coordinator with the Data Analysis and Cybersecurity research team. His research interests revolve around applications of machine learning to real world problems, and specially network security. He has published more than 25 articles in high impact journals, more than 44 publications in blind peer-reviewed conferences, edited 3 books (2 of them as conference proceedings), and coauthored 9 book chapters. He serves as Program Committee member in conferences such as RAID, DIMVA or SECRIPT, and as Steering Committee member in RAID and DIMVA.



Mauro Conti (Fellow, IEEE) received the PhD degree from the Sapienza University of Rome, Italy, in 2009. He is a full professor with the University of Padua, Italy. He is also a Wallenberg-WASP guest professor Örebro University. After his PhD, he was a post-doc researcher with Vrije Universiteit Amsterdam, The Netherlands. In 2011 he joined as assistant professor with the University of Padua, where he became an associate professor in 2015, and a full professor in 2018. He has been visiting researcher with GMU, UCLA, UCI, TU Darmstadt, UF, and FIU. He has been awarded with a Marie Curie Fellowship (2012) by the European Commission, and with a Fellowship by the German DAAD (2013). His research is also funded by companies, including Cisco, Intel, and Huawei. His main research interest is in the area of security and privacy. In this area, he published more than 550 papers in topmost international peer-reviewed journals and conferences. He is a editor-in-chief for *IEEE Transactions on Information Forensics and Security*, area editor-in-chief for *IEEE Communications Surveys & Tutorials*, and has been associate editor for several journals, including *IEEE Communications Surveys & Tutorials*, *IEEE Transactions on Dependable and Secure Computing*, *IEEE Transactions on Information Forensics and Security*, and *IEEE Transactions on Network and Service Management*. He was program chair for TRUST 2015, ICISS 2016, WiSec 2017, ACNS 2020, CANS 2021, CSS 2021, WiMob 2023 and ESORICS 2023, and general chair for SecureComm 2012, SACMAT 2013, NSS 2021 and ACNS 2022. He is fellow of the AAIA, distinguished member of the ACM, and fellow of the Young Academy of Europe.